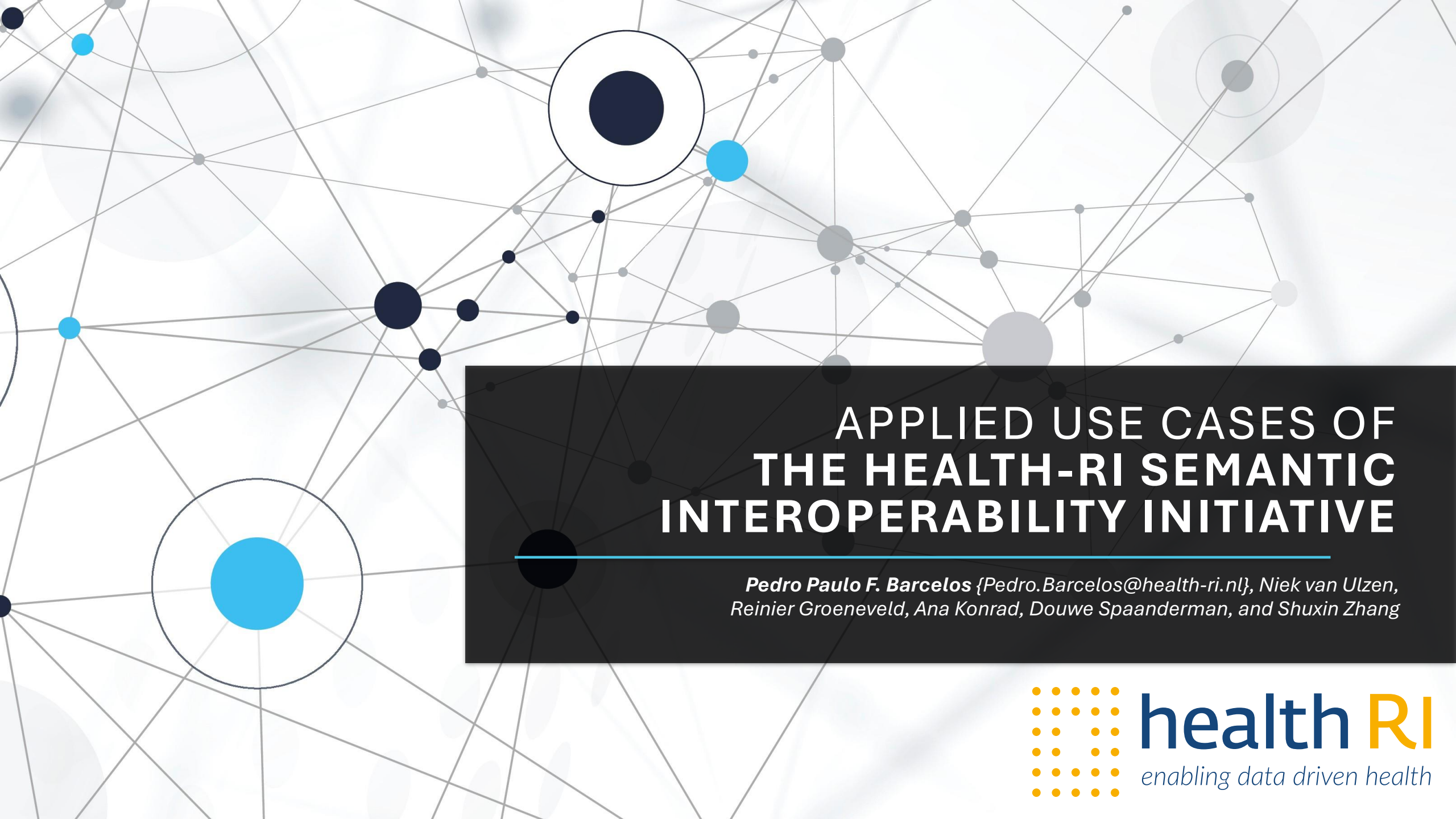




APPLIED USE CASES OF THE HEALTH-RI SEMANTIC INTEROPERABILITY INITIATIVE

*Pedro Paulo F. Barcelos {Pedro.Barcelos@health-ri.nl}, Niek van Ulzen,
Reinier Groeneveld, Ana Konrad, Douwe Spaanderman, and Shuxin Zhang*

AMIR HASSAN: RESEARCH GOAL AND CHALLENGE



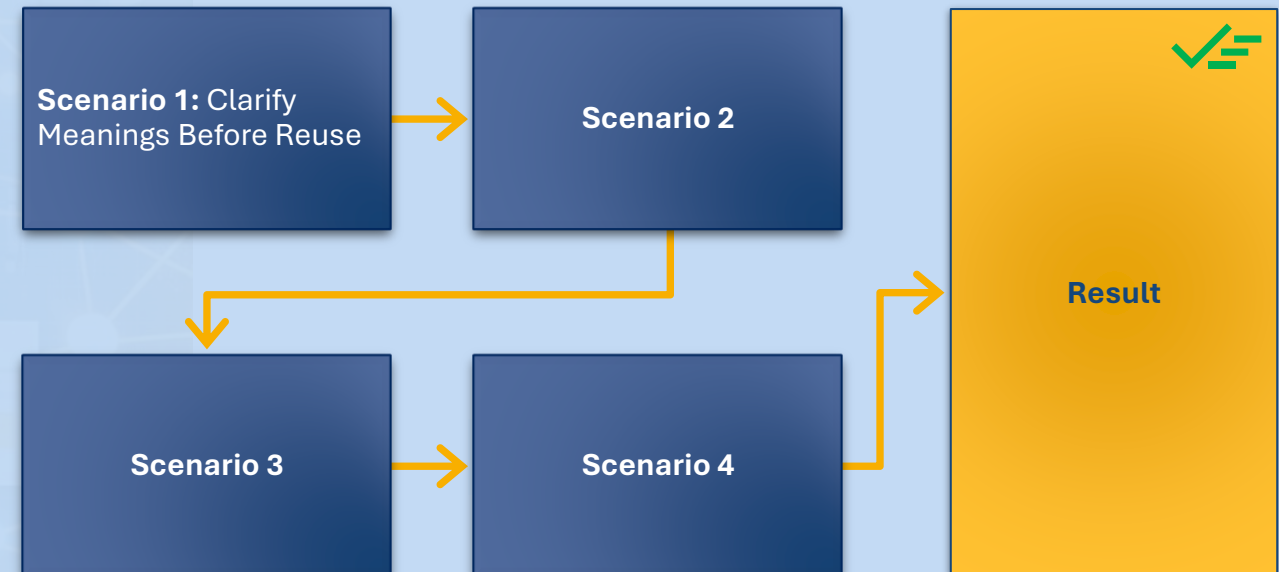
Dr. Amir Hassan — Researcher

***Amir** is studying whether treatment outcomes differ across patient groups and needs to stratify his cohort using the correct sex/gender-related variable.*

Before requesting data, he must identify which datasets contain the right variable for his study.



SCENARIO 1: CLARIFY MEANINGS BEFORE REUSE



SCENARIO 1: AMIR'S QUESTION AND CHALLENGE



What Amir wants

- Use the correct sex/gender-related variable to create a cohort that can support trustworthy analysis.

Amir's challenge

- Similar-looking variables across datasets may represent different meanings, making reuse risky.

Guiding question

- How can Amir avoid risky reuse when similar-looking variables may represent different meanings?



AVAILABLE SEX AND GENDER VARIABLES BY DATASET

NPR Demographic Extract				 CARE-SM Clinical And Registry Entries Semantic Model
patient_id	care-sm:Sex ("Biological sex"@en)	birth_year	residence_code	
NPR-001	female	1987	0363	
NPR-002	male	1979	0599	
NPR-003	female	1994	0518	

**National Health Data Authority**

Hospital Registration Extract				
person_id	hri-openehr:legal-gender ("Legal gender"@en)	encounter_id	record_date	
NMC-001	female	ENC-7001	2026-01-14	
NMC-002	male	ENC-7002	2026-01-15	
NMC-003	nonbinary	ENC-7003	2026-01-16	

**Northgate Medical Center**

Administrative Intake Extract				
intake_id	loinc:107454-1 ("Administrative gender"@en)	questionnaire_id	submitted_date	
RPCC-001	woman	GIQ-001	2026-02-03	
RPCC-002	man	GIQ-001	2026-02-03	
RPCC-003	non-binary	GIQ-002	2026-02-04	

**Riverside Primary Care Clinic**



LABELS AND THEIR MEANINGS

patient_id	CARE-SM:Sex ("Biological sex"@en)	birth_year	res
NPR-001	female	1987	036
NPR-002	male	1979	059
NPR-003	female	1994	051

Same term. Do they share the same meaning?

Phic Extract

al Health Authority



CARE-SM
Clinical And Registry Entries
Semantic Model

person_id	hri-openehr:legal-gender ("Legal gender"@en)	encounter_id	rec
NMC-001	female	ENC-7001	202
NMC-002	male	ENC-7002	202
NMC-003	nonbinary	ENC-7003	202

What is the difference between 'man' and 'male'?

Registration Extract

hgate Center



open EHR

intake_id	loinc:107454-1 ("Administrative gender"@en)	questionnaire_id	submitted_date
RPCC-001	woman	GIQ-001	2026-02-03
RPCC-002	man	GIQ-001	2026-02-03
RPCC-003	non-binary	GIQ-002	2026-02-04

Administrative Intake Extract



Riverside
Primary Care Clinic





SIMILAR LABELS CAN MISLEAD REUSE



Patient-001

(Biological) Sex: Male
(Social) Gender: Male



Patient-002

(Biological) Sex: Male
(Social) Gender: Female (Trans)

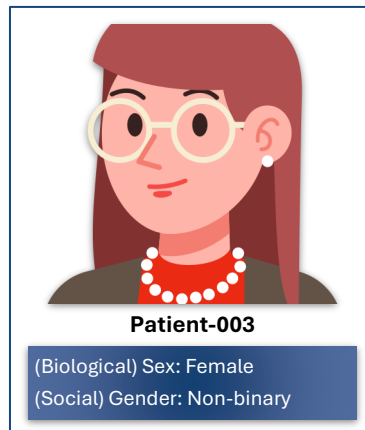
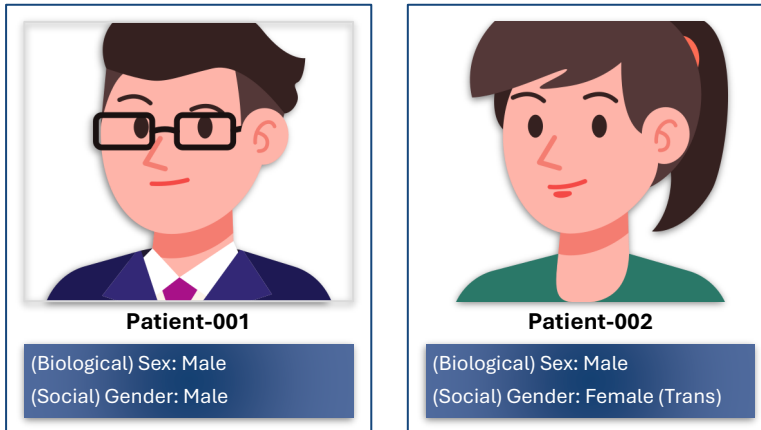


Patient-003

(Biological) Sex: Female
(Social) Gender: Non-binary



SIMILAR LABELS CAN MISLEAD REUSE



Identify patients with male biological sex for a study where biological sex is the intended criterion.

Retrieve patients where Gender = Male.

Returns Patient-001 only, incorrectly missing Patient-002, whose biological sex is male but gender is female/trans.

Identify patients who socially identify as male for gender-based analysis.

Retrieve patients where Sex = Male.

Returns Patient-001 and Patient-002, incorrectly including Patient-002, whose gender is female/trans.

Identify patients with female gender for a study about social gender identity.

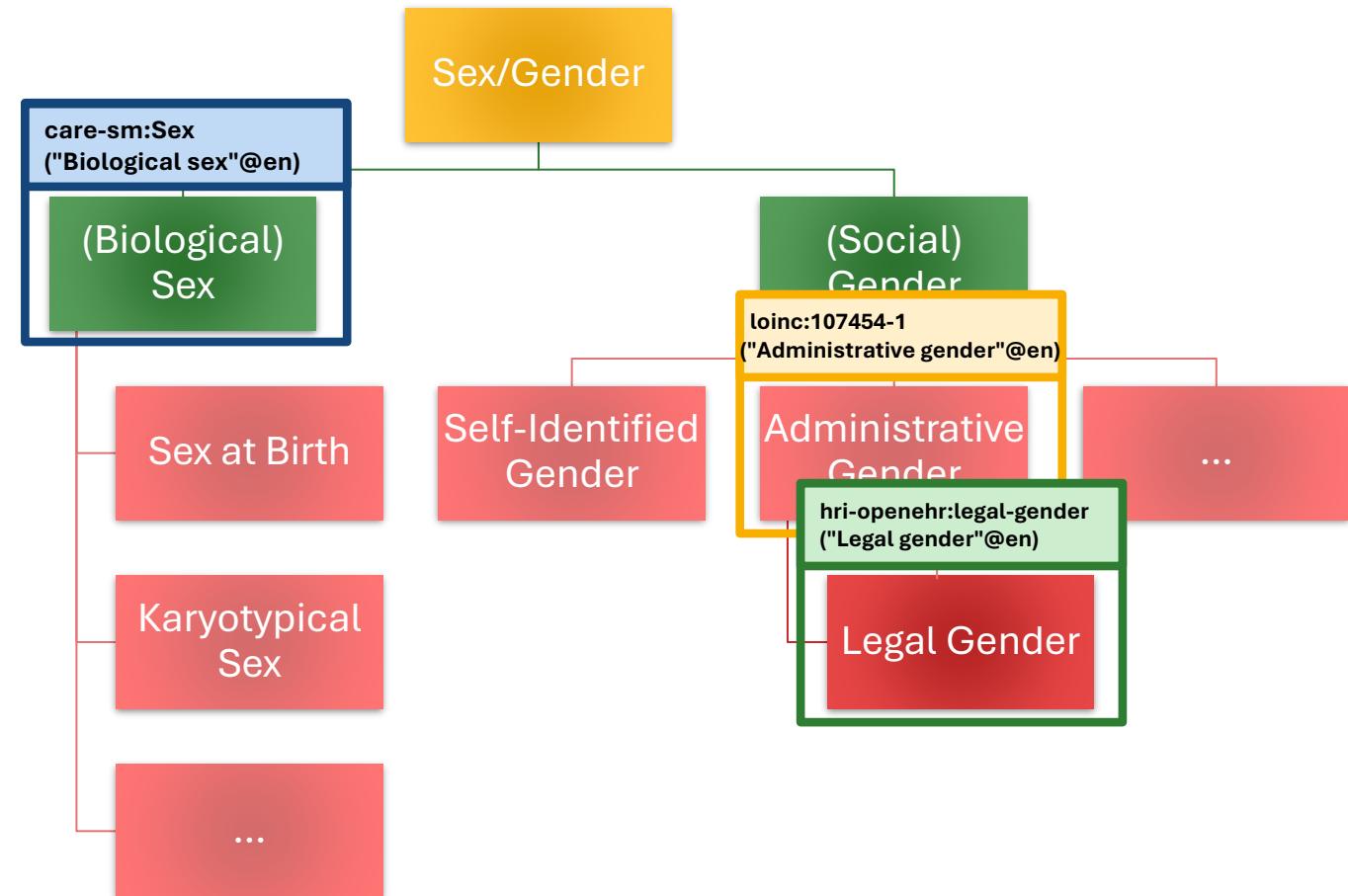
Retrieve patients where Sex = Female.

Returns Patient-003 only, incorrectly missing Patient-002, whose gender is female.



DISTINCT CONCEPTS WITHIN SEX AND GENDER

- Sex and Gender are not single, uniform meanings; each branches into multiple interpretations/specializations.
- Before comparing dataset variables, we must distinguish which specialization a concept actually refers to.

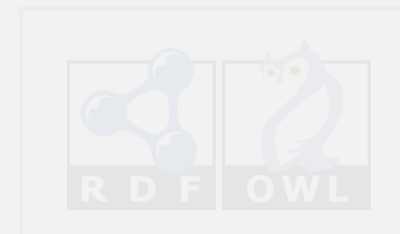
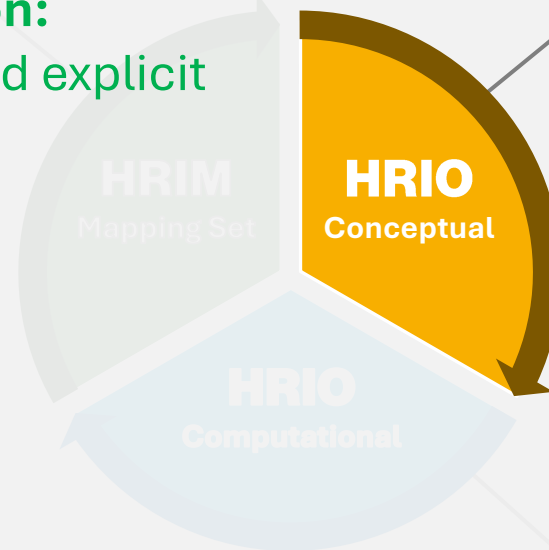


THE HEALTH-RI CONCEPTUAL ONTOLOGY

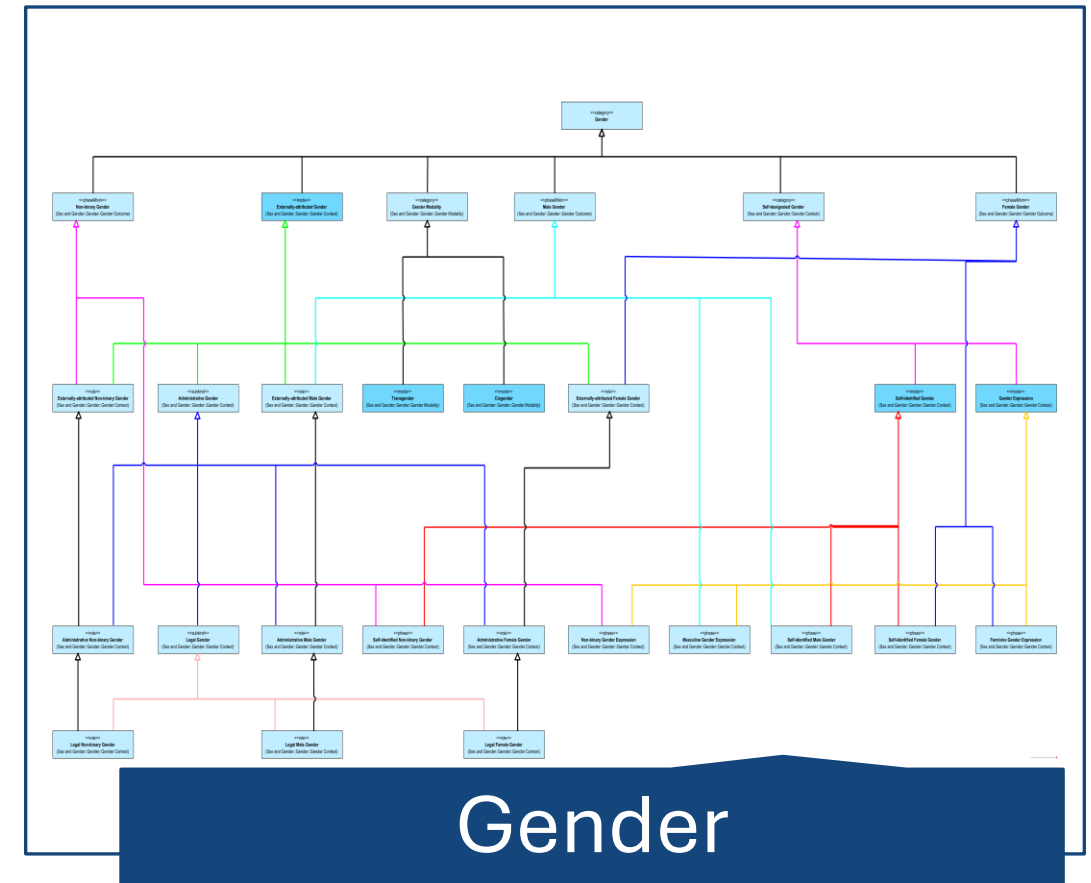
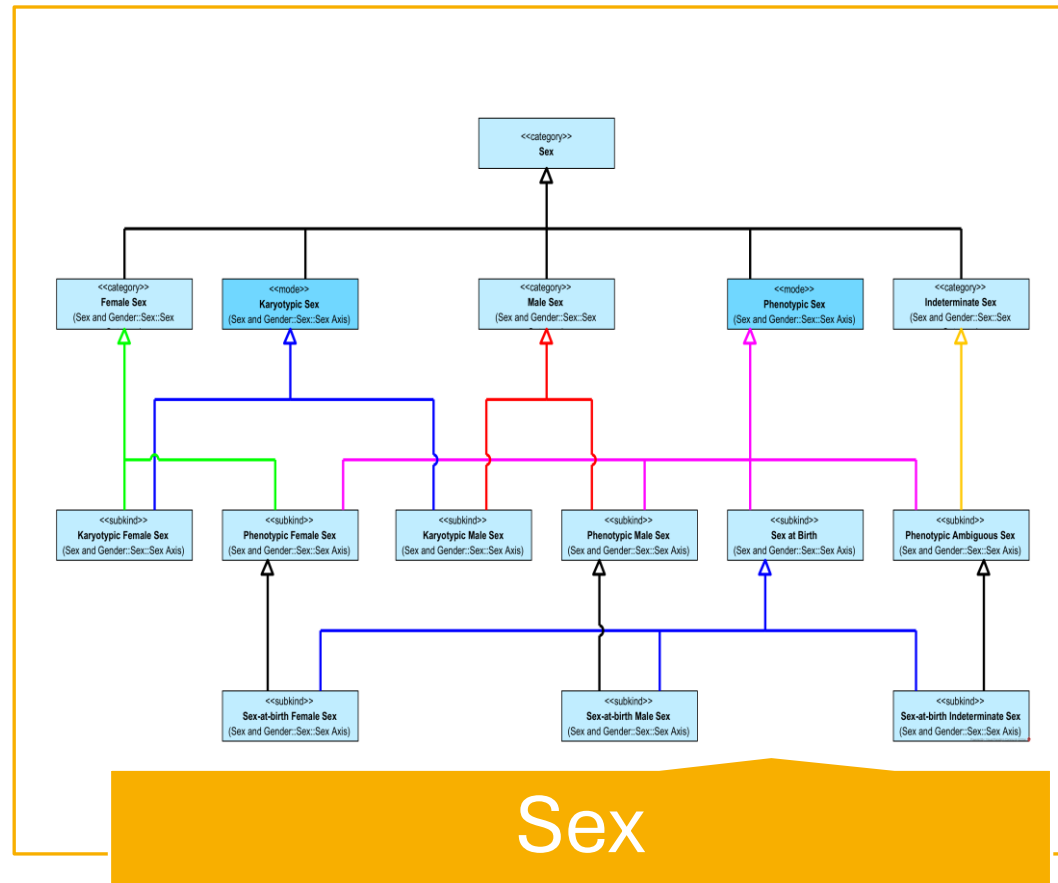
The Health-RI Semantic Interoperability Initiative



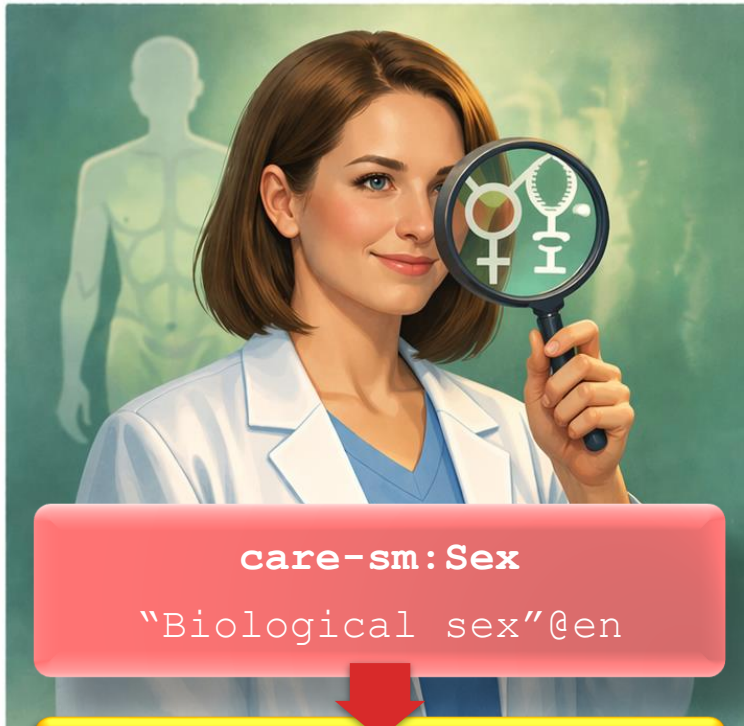
Conceptual clarification:
Meanings are precise and explicit



HRIO TAXONOMIES FOR SEX AND GENDER



VARIABLES MAP TO DISTINCT MEANINGS



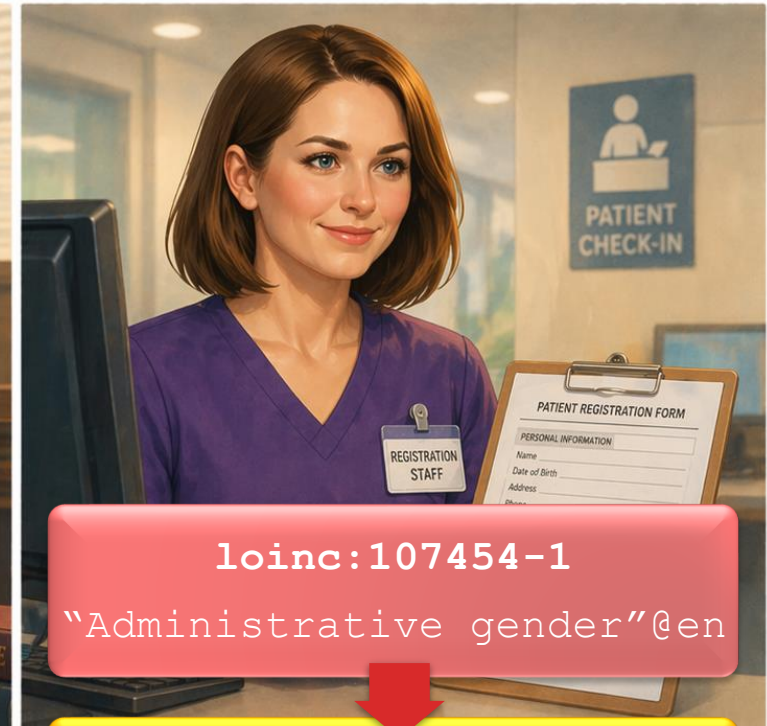
care-sm:Sex
"Biological sex"@en

HRIO's Sex



hri-openehr:legal-gender
"Legal gender"@en

HRIO's Legal Gender

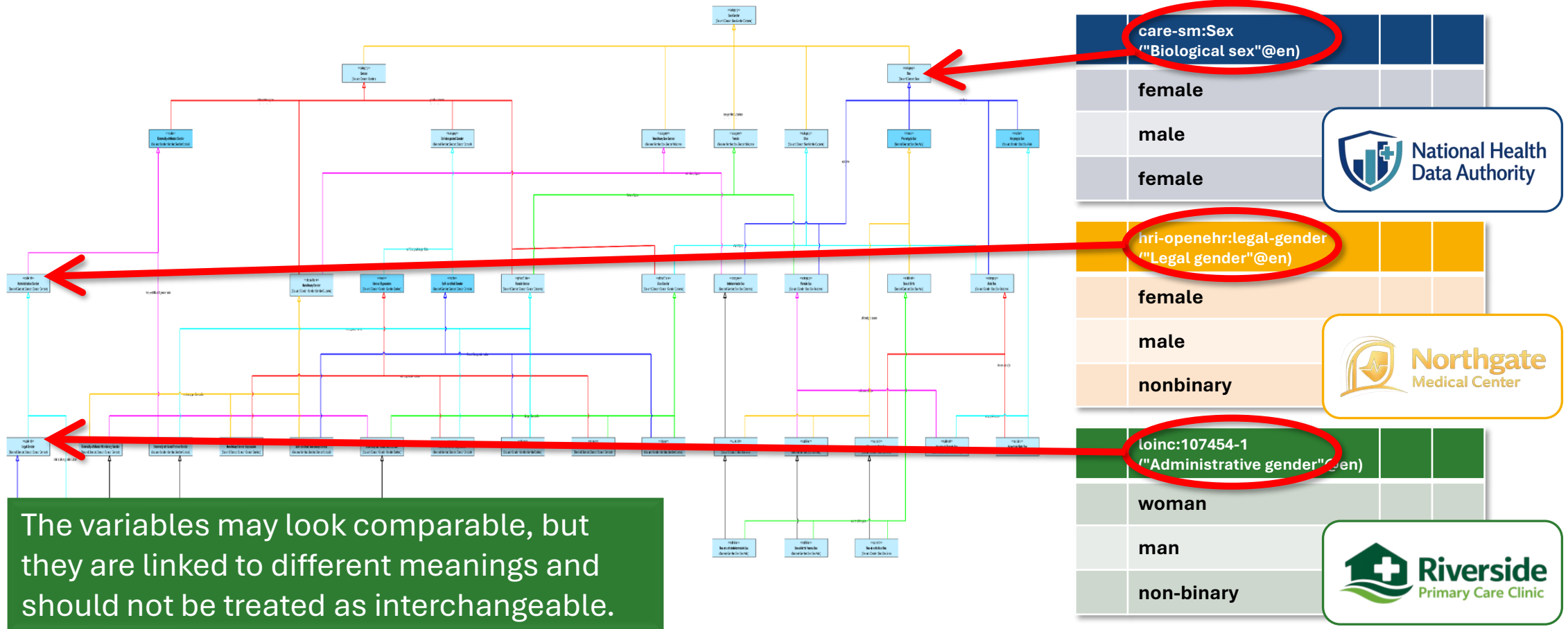


loinc:107454-1
"Administrative gender"@en

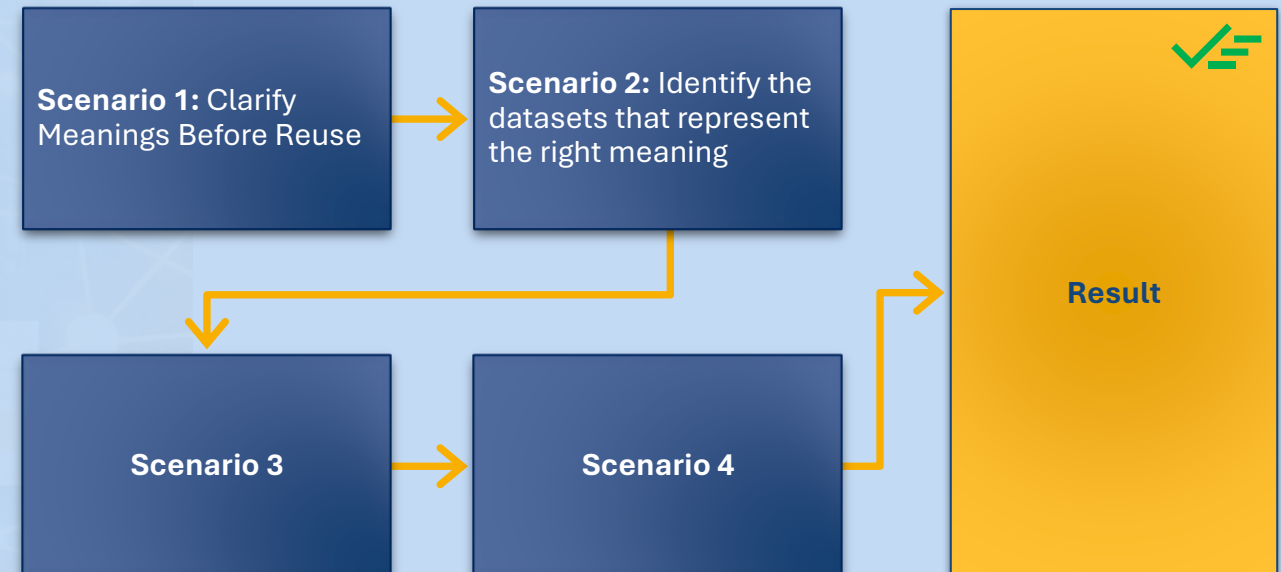
HRIO's Administrative Gender



SIMILAR VARIABLES, DIFFERENT MEANINGS



SCENARIO 2: IDENTIFY THE DATASETS THAT REPRESENT THE RIGHT MEANING



SCENARIO 2: AMIR'S QUESTION AND CHALLENGE



What Amir wants

- Find the dataset column for biological sex so he can build a cohort that fits his research question.

Amir's challenge

- Several datasets appear relevant, but their sex/gender-related columns represent different meanings, making label-based selection unsafe.

Guiding question

- Which dataset contains the variable that represents biological sex?

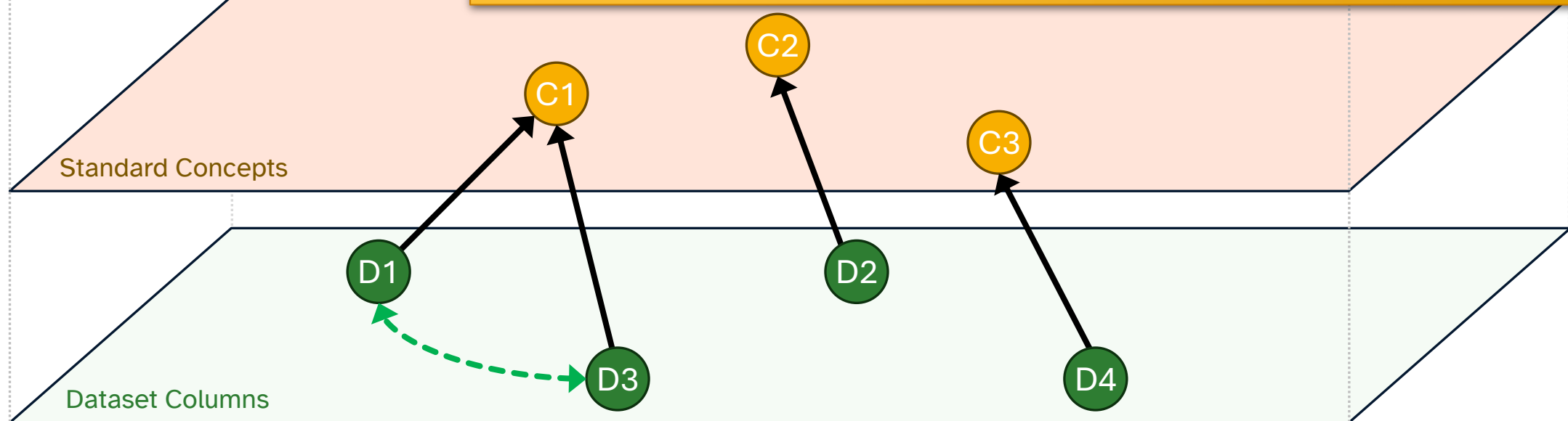


WHEN ARE TWO DATASETS INTEROPERABLE?

When can two datasets be safely reused together?

When their relevant columns implement the same standard concept.

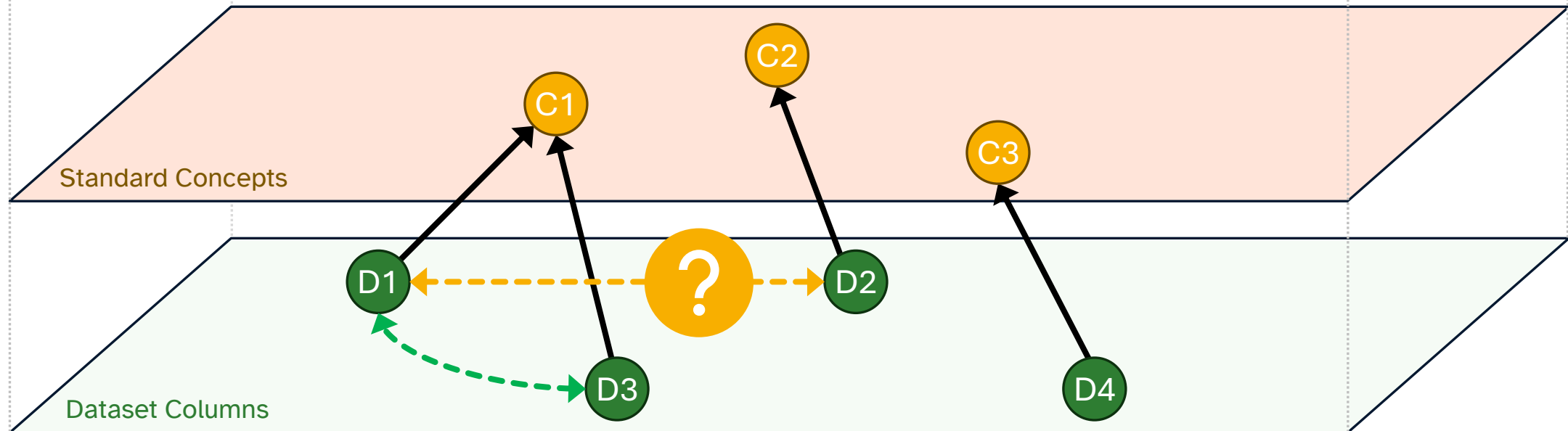
However, this works only if (1) the standard concept is well defined, and (2) the datasets implement it correctly.



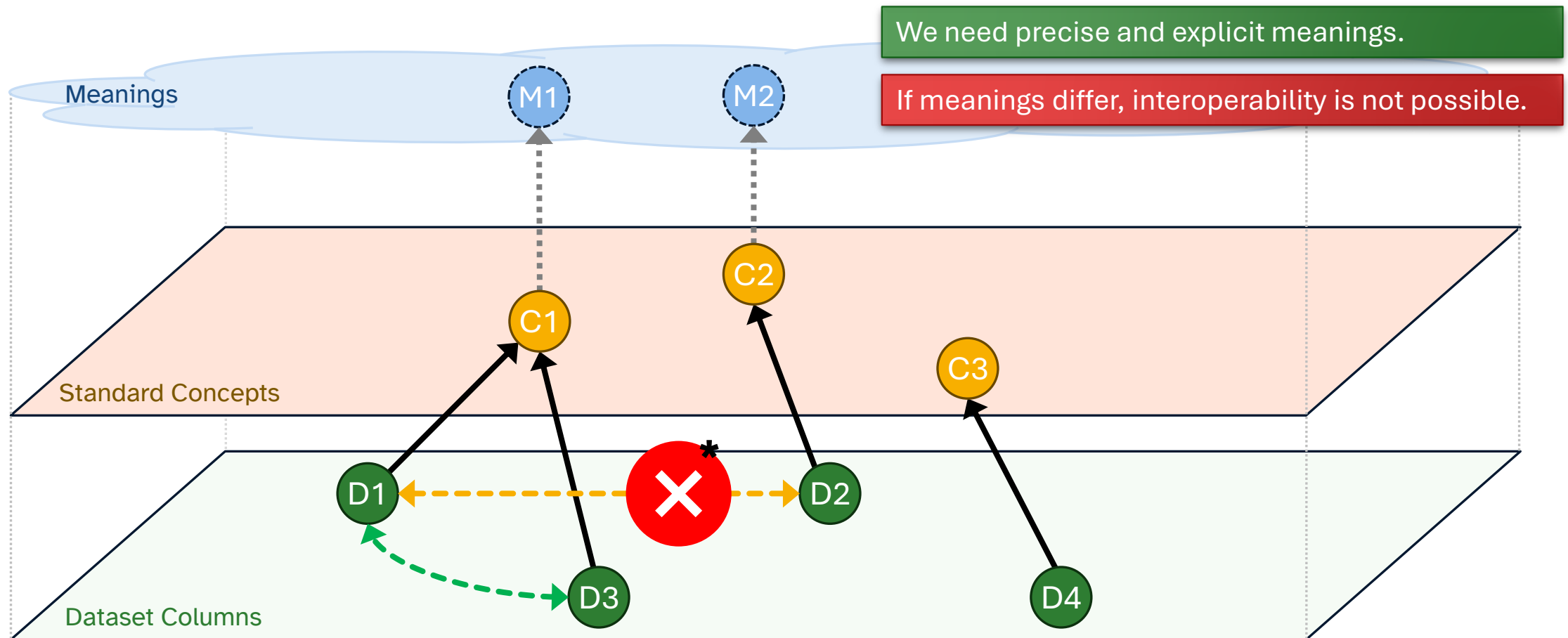
WHEN ARE TWO DATASETS INTEROPERABLE?

Can **D1** and **D2** be safely reused together?

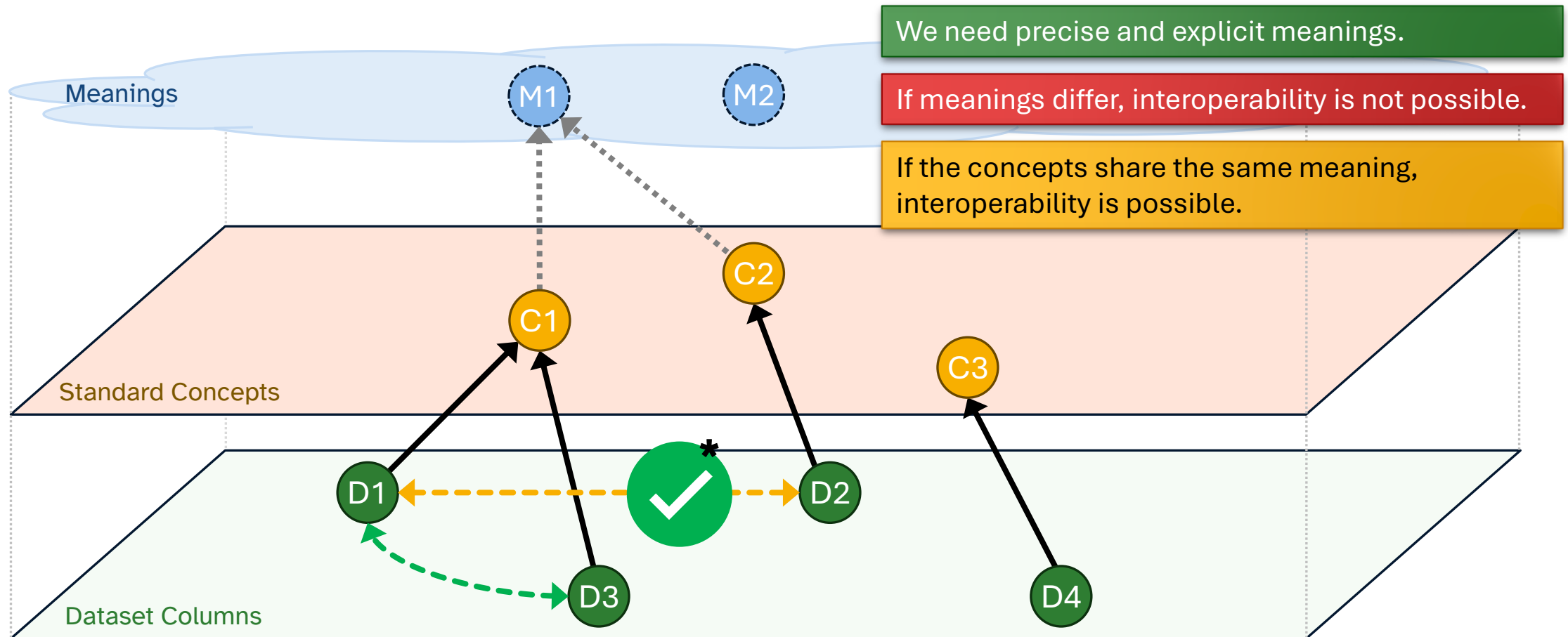
Without explicit meanings, we can only see interoperability when datasets share the same standard concept.



SEMANTIC INTEROPERABILITY CONDITIONS

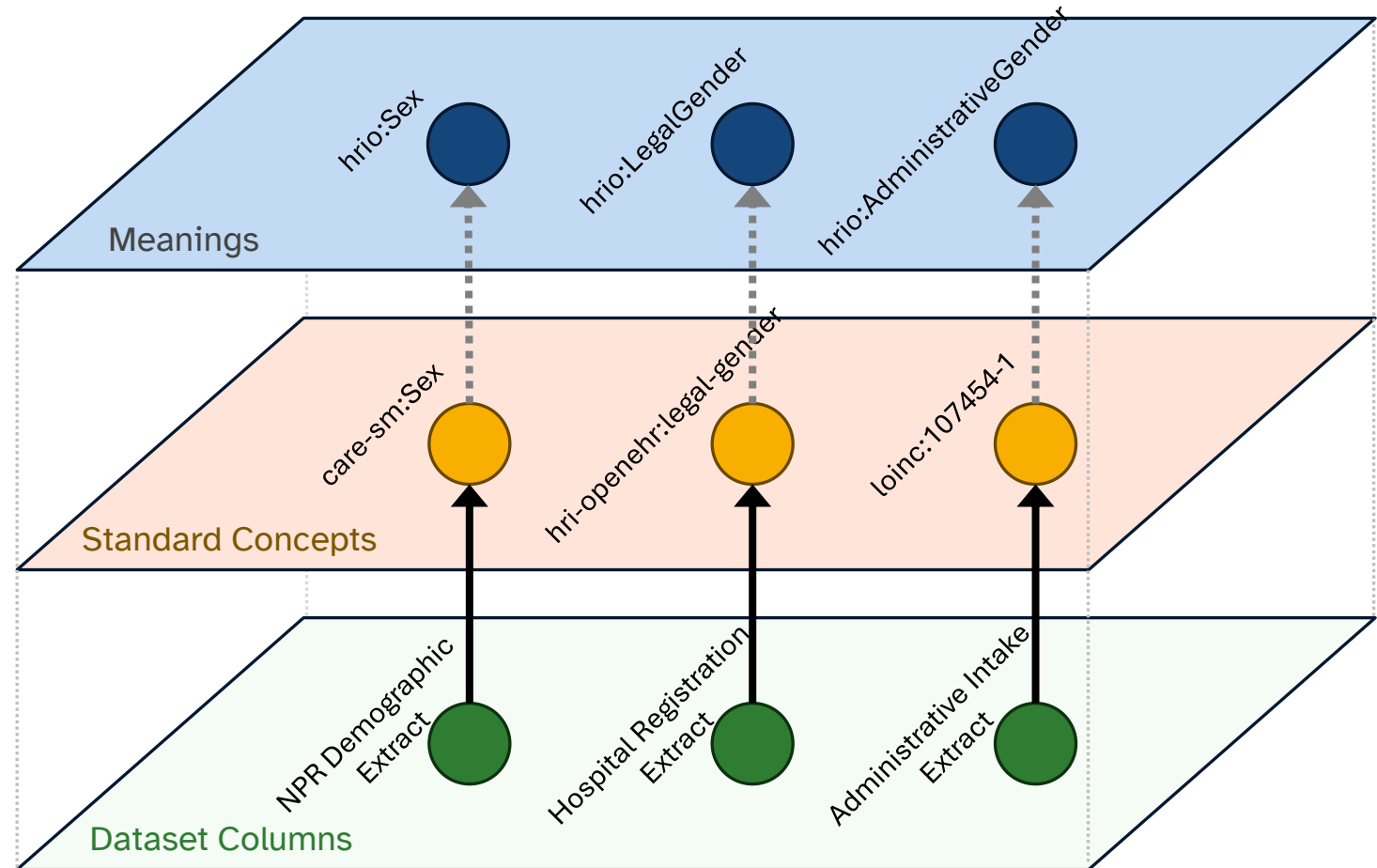


SEMANTIC INTEROPERABILITY CONDITIONS



FROM CONCEPTUAL CLARITY TO ACTIONABLE SEMANTICS

- The meanings are now clear in the conceptual model, but Amir still cannot query or compute over them directly.
- The next step is to turn these conceptual meanings into machine-readable, queryable, and actionable relations.



THE HEALTH-RI COMPUTATIONAL ONTOLOGY

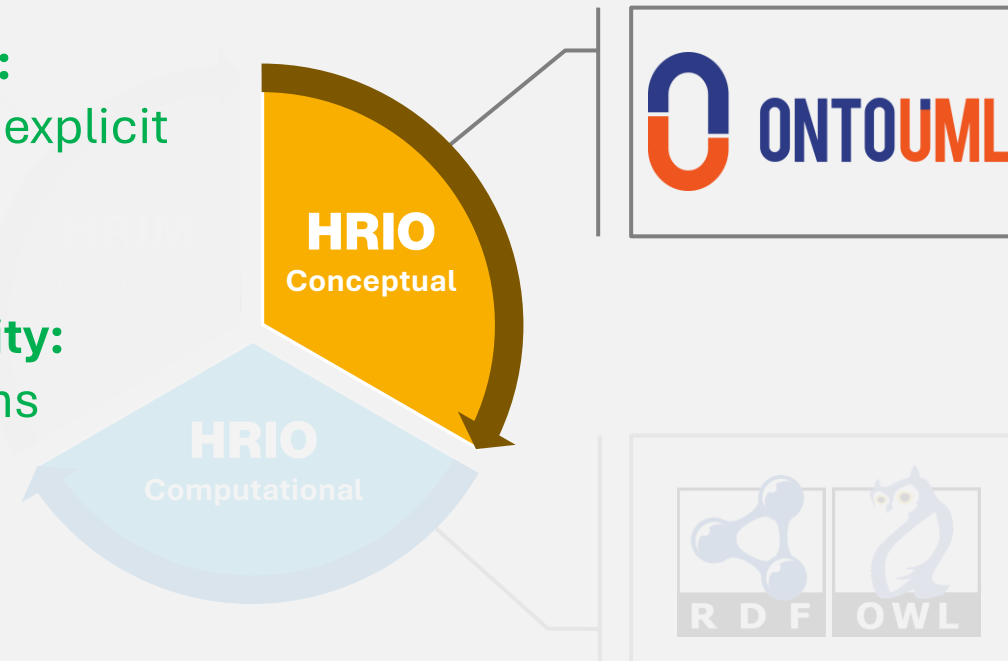
The Health-RI Semantic Interoperability Initiative



Conceptual clarification:
Meanings are precise and explicit



Computational tractability:
Machine-readable relations



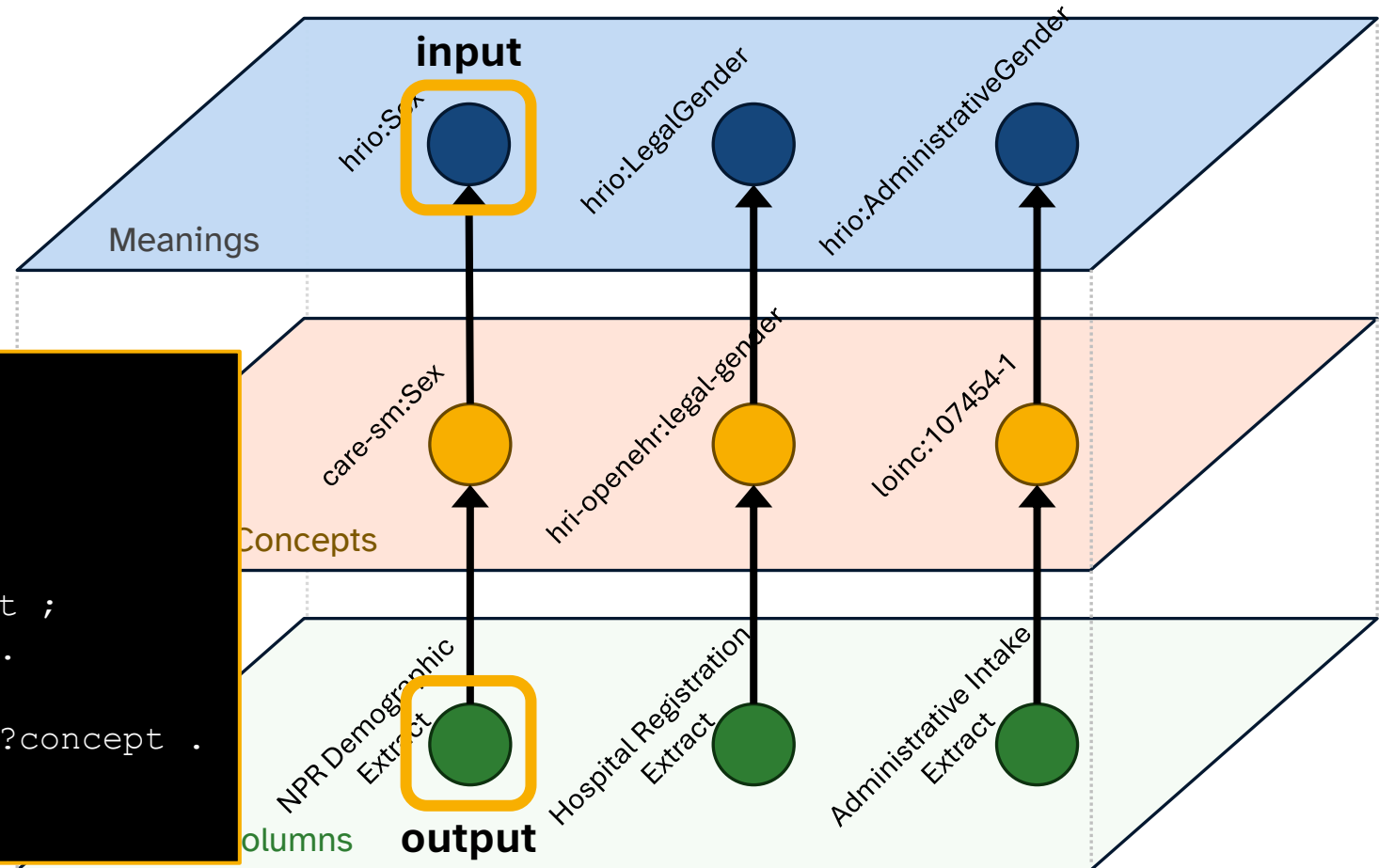
FROM CONCEPTUAL CLARITY TO ACTIONABLE SEMANTICS

- Amir can now query for dataset columns that are mapped to the meaning he needs: HRIO's Sex.

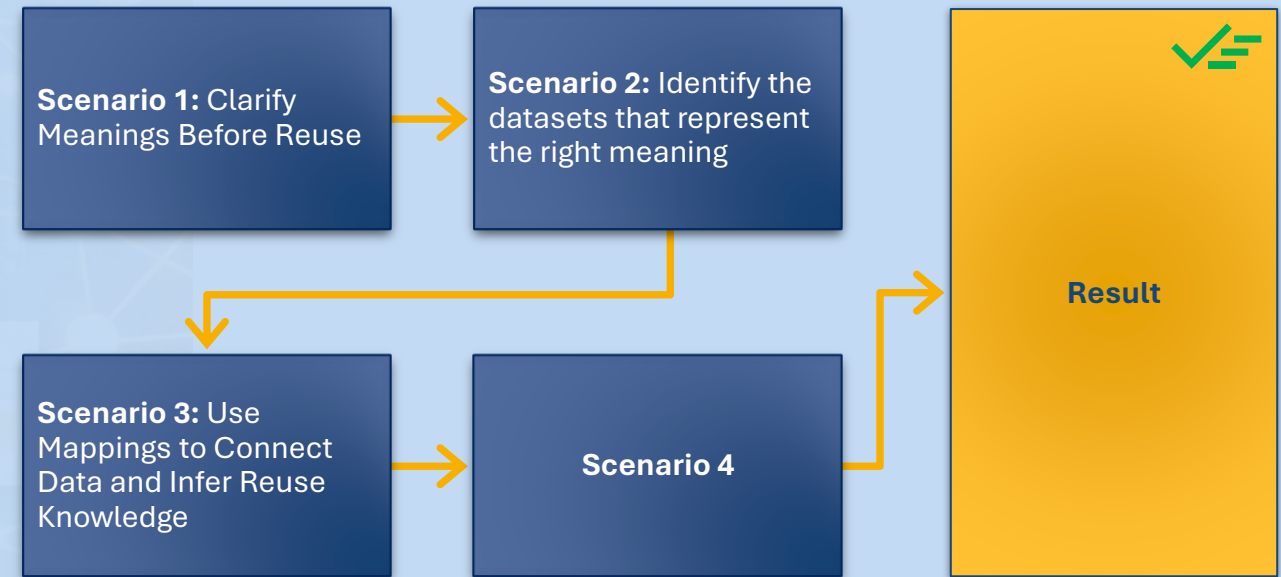
```
SELECT DISTINCT ?output
WHERE {
  VALUES ?input { hrio:Sex }

  ?mapping a demo:Mapping ;
    demo:hasMappingSource ?concept ;
    demo:hasTargetMeaning ?input .

  ?output demo:implementsConcept ?concept .
}
ORDER BY ?output
```



SCENARIO 3: USE MAPPINGS TO CONNECT DATA AND INFER REUSE KNOWLEDGE



SCENARIO 3: AMIR'S QUESTION AND CHALLENGE



What Amir wants

- Apply reusable mappings to relate dataset variables, standard concepts, and explicit meanings.

Amir's challenge

- The meanings are explicit and machine-readable, but these relationships must be captured as reusable mappings.

Guiding question

- How can mappings turn these relationships into reusable knowledge?



EXTERNAL MAPPINGS

Clinical Research Standards Mapping Community

source_CURIE	source_label	target_CURIE	target_label
cdisc-sdtm:DM.SEX	Sex	care-sm:Sex	Biological sex
ncit:C28421	Sex	care-sm:Sex	Biological sex

Dutch Health Data Stewardship Community

source_CURIE	source_label	target_CURIE	target_label
npr:person.sex_code	Registered biol. sex	care-sm:Sex	Biological sex
npr:person.sex_code	Registered biol. sex	registry:resident.sex	Resident sex
fhir:Patient.gender	Administrative gender	hl7v3:AdmGender	Administrative gender

Administrative Gender Standards Community

source_CURIE	source_label	target_CURIE	target_label
loinc:107454-1	Administrative gender	fhir:Patient.gender	Administrative gender
loinc:107454-1	Administrative gender	hl7v2:0001	Administrative sex

care-sm:Sex ("Biological sex"@en)		
female		
male		
female		



hri-openehr:legal-gender ("Legal gender"@en)		
female		
male		
nonbinary		



loinc:107454-1 ("Administrative gender"@en)		
woman		
man		
non-binary		



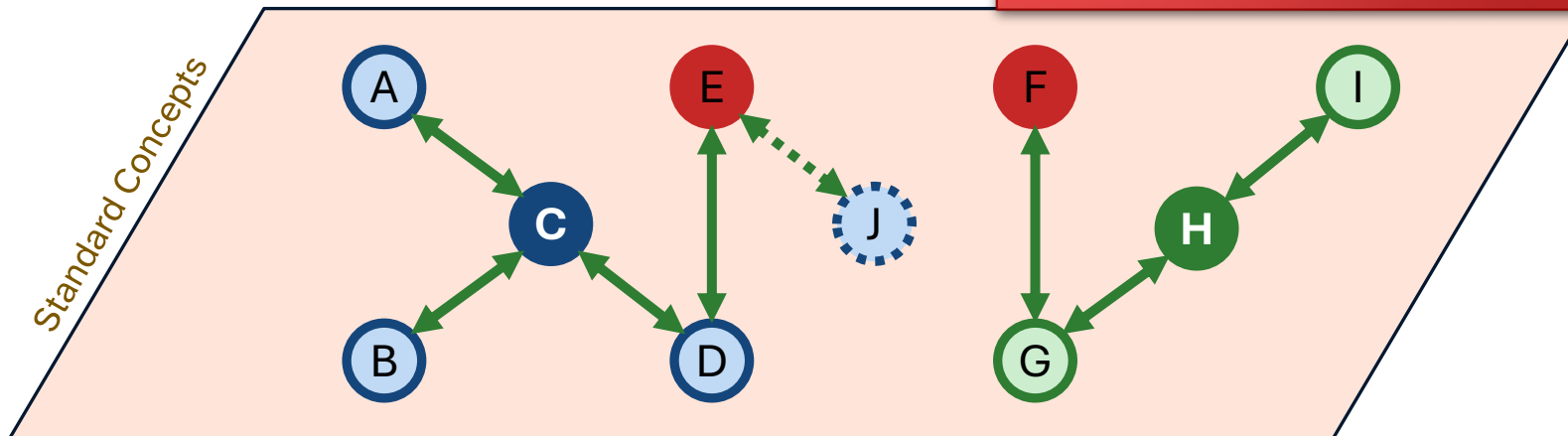
PAIRWISE MAPPINGS REQUIRE MULTIPLE STEPS

- Without a meaning layer, reuse depends on chains of concept-to-concept mappings.
- These mappings are often fragile and label-driven.
- Each new concept adds complexity, making maintenance harder.

From **E** to **A**: 3 steps

From **J** to **A**: 4 steps

From **Z** to **A**: N steps



source_CURIE	target_CURIE
cdisc-sdtm:DM.SEX A	care-sm:Sex C
ncit:C28421 B	care-sm:Sex

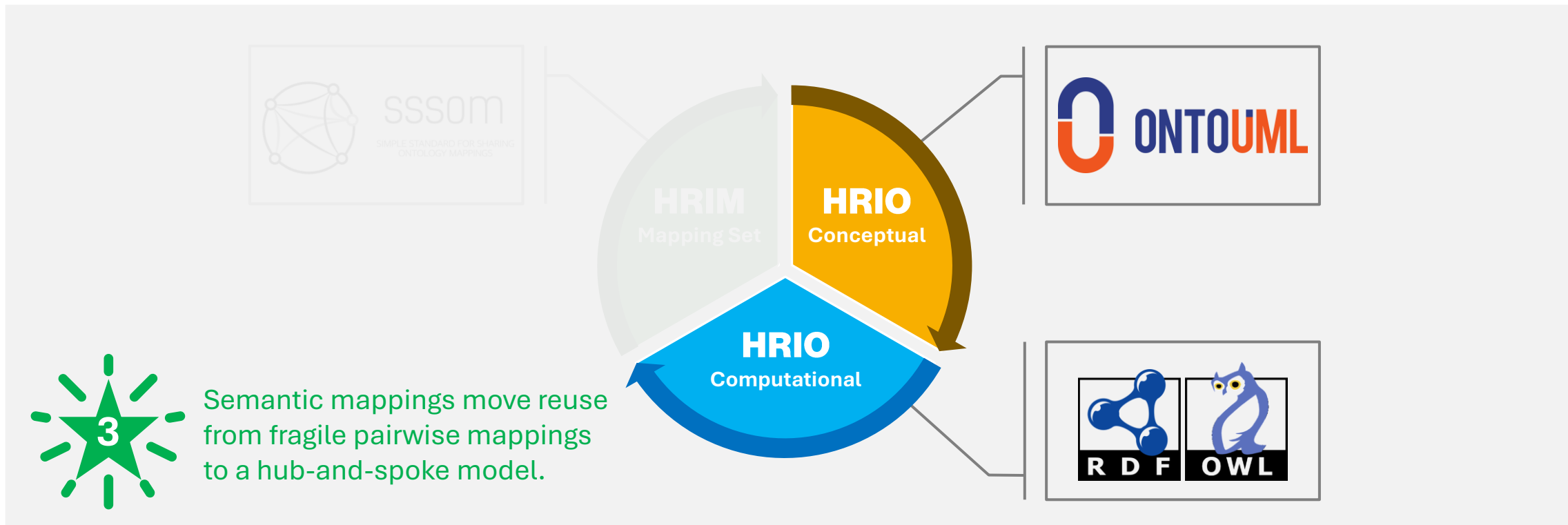
source_CURIE	target_CURIE
npr:person.sex_code D	care-sm:Sex
npr:person.sex_code	registry:resident.sex E
fhir:Patient.gender	hl7v3:AdmGender F

source_CURIE	target_CURIE
loinc:107454-1 H	fhir:Patient.gender G
loinc:107454-1	hl7v2:0001 I

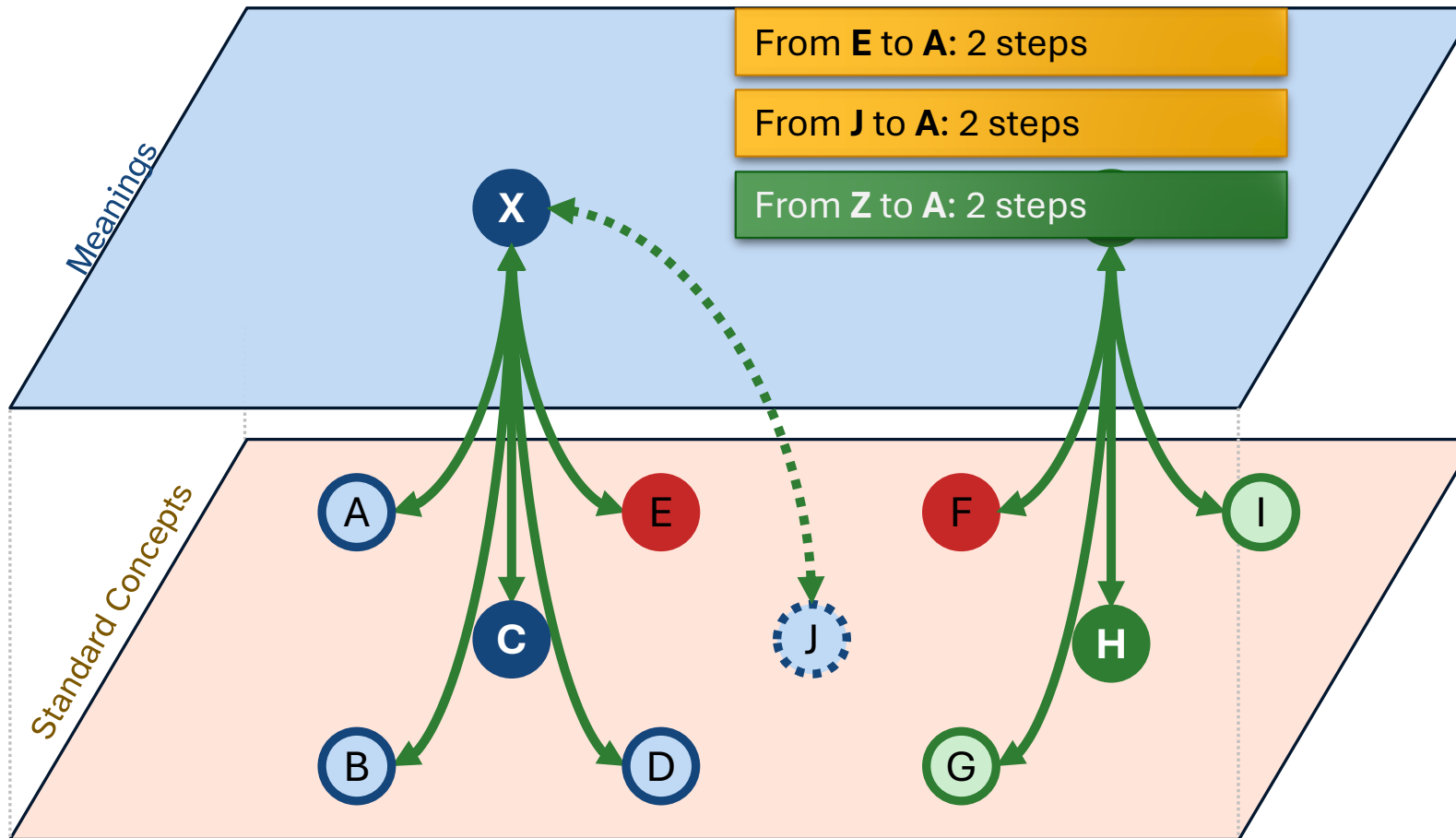


THE HEALTH-RI SSSOM MAPPING SET (HRIM)

The Health-RI Semantic Interoperability Initiative



MEANING-BASED MAPPINGS SHORTEN REUSE PATHS

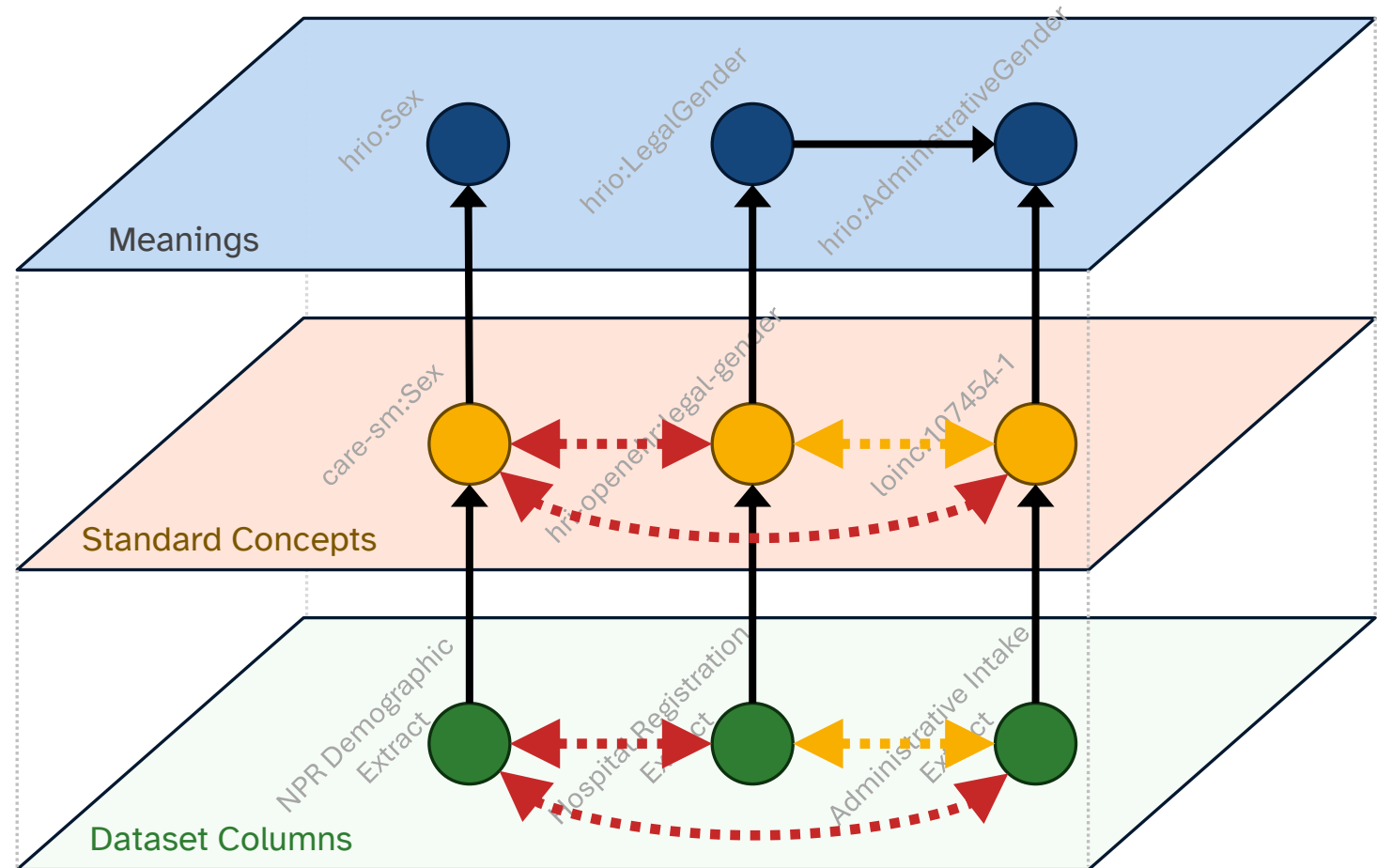


- Shared meanings create a common semantic hub. Paths are shorter and clearer.
- Semantic alignment is based on explicit meanings, not labels.
- Adding concepts creates more reusable knowledge for future queries, enriching the network without increasing path complexity.

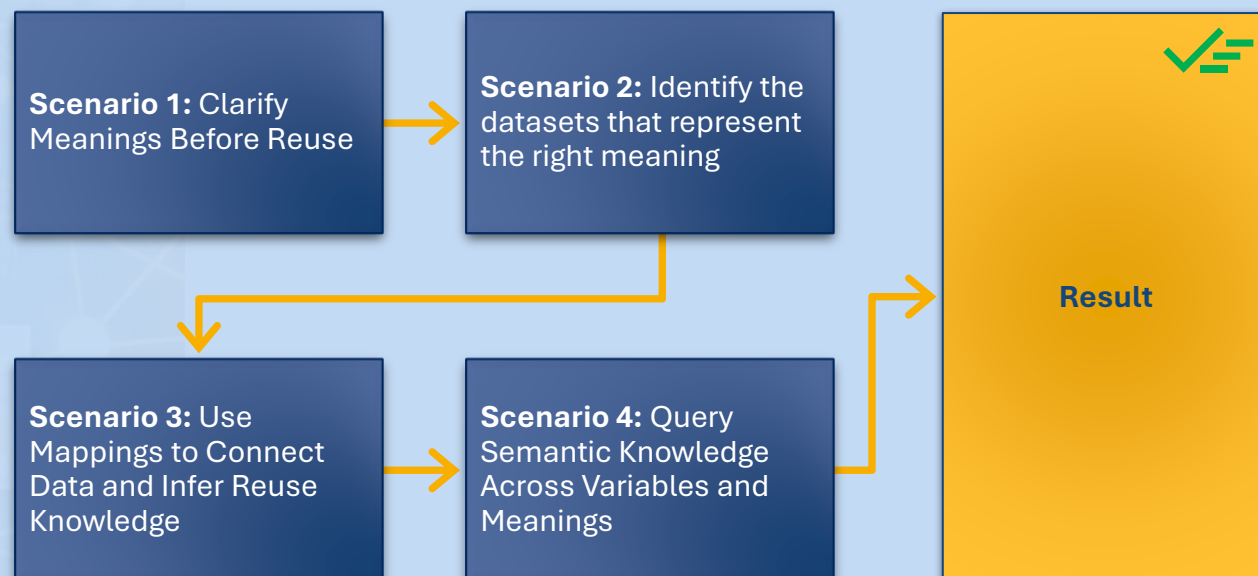


AMIR ASSESSES REUSE THROUGH SEMANTIC RELATIONS

- Biological sex does not align with the gender-based variables.
- Legal gender and administrative gender are related, but only partially align.
- Amir can now rule out unsafe reuse.



SCENARIO 4: QUERY SEMANTIC KNOWLEDGE ACROSS VARIABLES AND MEANINGS



SCENARIO 4: AMIR'S QUESTION AND CHALLENGE



What Amir wants

- Query semantic context, mappings, and inferred knowledge to support data reuse decisions.

Amir's challenge

- Amir needs to move from connected semantic knowledge to queries that support concrete reuse decisions.

Guiding question

- How can Amir query semantic knowledge to support practical data reuse decisions?

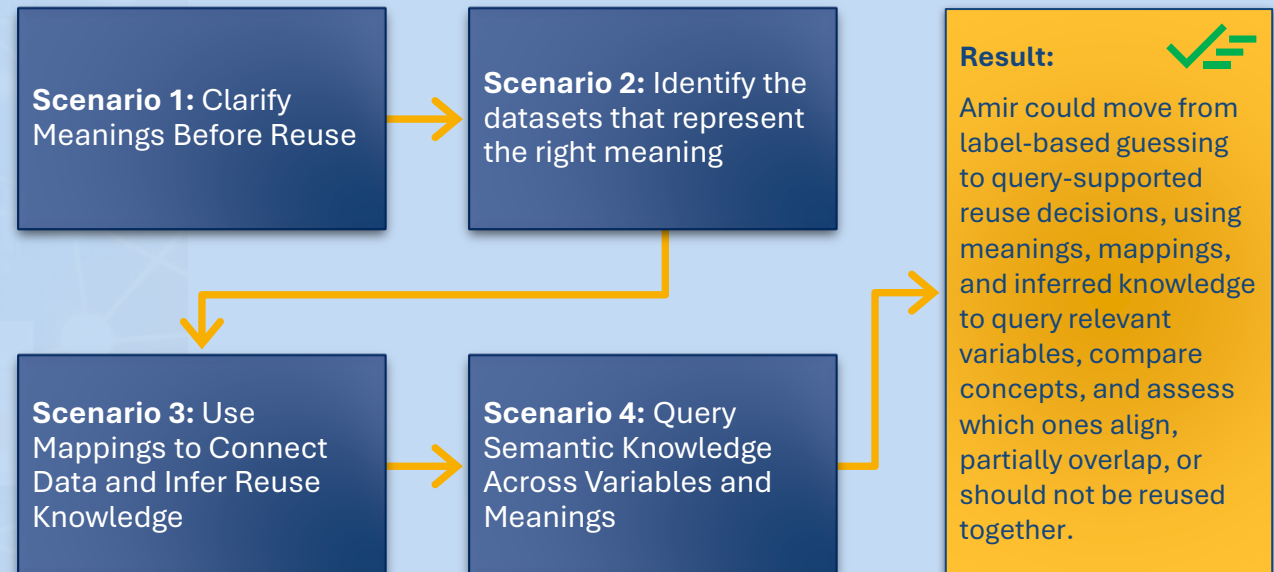


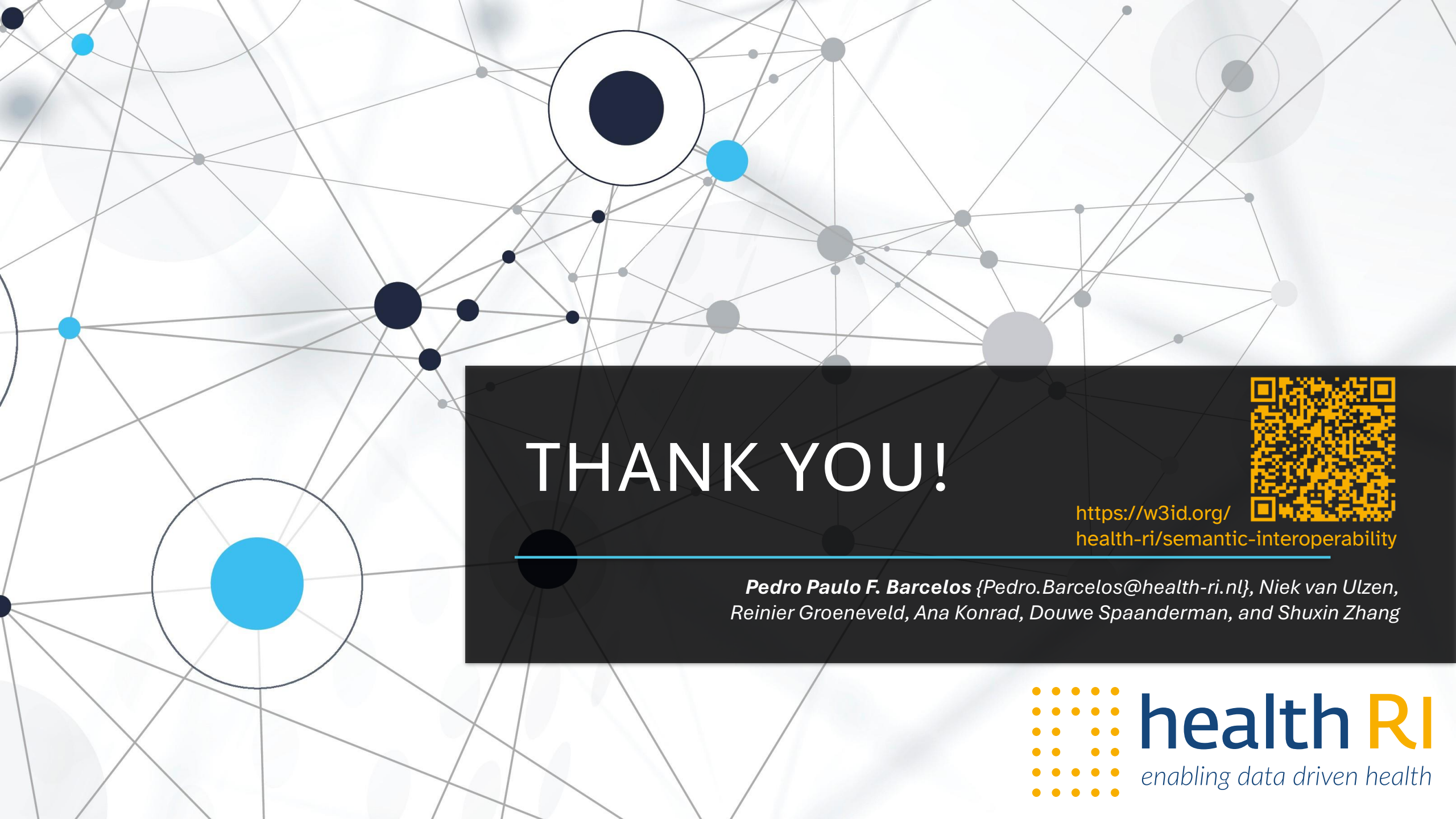
AMIR'S POSSIBLE QUERIES

A Discover what is available.			
Query		What Amir gains	
	What sex- and gender-related meanings are represented in the available datasets?		Amir sees which meanings are available to choose from.
	Which dataset variables are connected to each meaning?		Amir can find variables by meaning, not only by label.
	Which standard concepts are represented by each dataset variable?		Amir understands which standards each variable uses.
B Retrieve patient records with semantic context			
Query		What Amir gains	
	Return all patient records with that dataset variable, standard concept, and meaning.		Amir sees each record with the right context for his study.
	Return records with a selected standard value, together with the dataset variable, standard concept, and meaning.		Amir compares similar values, ensuring they mean the same thing.
	Return all patient records described by a gender-related meaning.		Amir can refine records by meaning, even when labels differ.
	Return all patient records described by a biological sex.		Amir can generate patient-level tables for his study.
C Assess semantic compatibility			
Query		What Amir gains	
	Are variable D1 and variable D2 semantically compatible for Amir's study?		Amir receives clear guidance for comparing two variables.
	Which dataset variables align with the meaning used by D1?		Amir finds safer reuse conditions.
	Which dataset variables partially overlap with the meaning used by D1?		Amir finds possible conflicts for his analysis.
	Which dataset variables should be ruled out for reuse with D1?		Amir avoids misusing variables with incompatible meanings.
D Use inferred reuse knowledge			
Query		What Amir gains	
	Which reuse relation can be inferred from the available mappings?		Amir discovers reuse relations that are not stated directly.
	Which concepts become connected through shared meanings?		Amir sees how his variables may become comparable.
	Which additional dataset variables become discoverable through indirect or inferred mappings?		Amir finds relevant variables he might otherwise miss.



RESULT OF AMIR'S WORKFLOW



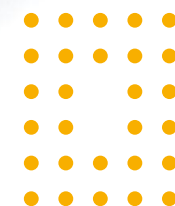


THANK YOU!



[https://w3id.org/
health-ri/semantic-interoperability](https://w3id.org/health-ri/semantic-interoperability)

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enabling data driven health